

do-while Loop (10 Problems)

1.Print numbers from 1 to 10

```
#include <iostream>
using namespace std;
//1.Print numbers from 1 to 10
int main(){
    int num = 1;
    do{
        cout << num <<" ";
        num++;
    }while(num <= 10);

    return 0;
}
```

2.Print numbers from 10 to 1

```
#include <iostream>
using namespace std;

//Print numbers from 10 to 1
int main(){

    int num = 10;
    do{
        cout <<num <<" ";
        num--;
    }while(num > 0);

    return 0
}
```

3.Menu-driven calculator (+, -, *, /)

```
#include <iostream>
#include<limits>
using namespace std;

//Menu-driven calculator (+, -, *, /)
int main(){
    int choice;
```

```
double num1,num2;

do{
    cout <<"\n-----Calculation-----" <<endl;

    cout <<"Enter:-1 for Addition(+)" <<endl;
    cout <<"Enter:-2 for Subtraction(-)" <<endl;
    cout <<"Enter:-3 for Multiplication(*)" <<endl;
    cout <<"Enter:-4 for Division(/)" <<endl;
    cout <<"Enter:-5 for Exit Code." <<endl;
    cout <<"-----" <<endl;

    cout <<"Enter your Choice: ";

    while(!(cin >> choice)){
        cout <<"Invalid Input!"<<endl;
        cout <<"Enter again(1 to 5): ";

        cin.clear();
        cin.ignore(numeric_limits<streamsize>::max(), '\n');
    }

    if (choice >= 1 && choice <=4){
        cout <<"Enter First Number: ";
        while(!(cin >> num1)){
            cout <<"Invalid Input!" <<"Enter a Valid Number: ";

            cin.clear();
            cin.ignore(numeric_limits<streamsize>::max(), '\n');
        }
        cout <<"Enter a Second Number:" ;
        while(!(cin >> num2 )){
            cout <<"Invalid Input!" <<"Enter a Valid Number: ";

            cin.clear();
            cin.ignore(numeric_limits<streamsize>::max(), '\n');
        }
        switch (choice)
        {
            case 1:
                cout<< "Addition: " <<num1 + num2 <<endl;
                break;

            case 2:
                cout <<"Substraction: " <<num1 - num2 <<endl;
                break;

            case 3:
                cout <<"Multiplication: " <<num1 * num2 <<endl;
                break;
            case 4:
                if(num2 == 0){
                    cout <<"Error!" <<endl;
                }else{
```

```

        cout <<"Division: " << num1 / num2 <<endl;
    }
    break;
}
}
else if (choice == 5){
    cout <<"Thank You!" <<endl;
}else{
    cout <<"Invalid Choice!!!" <<endl;
}
}while(choice != 5);
return 0;
}

```

4.ATM simulation.\

```

#include <iostream>
#include <limits>
using namespace std;

int main() {
    double balance = 5000.0;
    int choice;

    do {
        cout << "\n===== \n";
        cout << "          ATM SIMULATION\n";
        cout << "===== \n";
        cout << "Current Balance: Rs. " << balance << endl;
        cout << "----- \n";
        cout << "1. Check Balance\n";
        cout << "2. Deposit Money\n";
        cout << "3. Withdraw Money\n";
        cout << "4. Exit\n";
        cout << "----- \n";
        cout << "Enter your choice: ";

        while (!(cin >> choice)) {
            cout << "Invalid input! Enter a number: ";
            cin.clear();
            cin.ignore(numeric_limits<streamsize>::max(), '\n');
        }

        switch (choice) {

            case 1:
                cout << "\nCurrent Balance = Rs. " << balance << endl;
                break;

            case 2: {

```

```
double deposit;

cout << "Enter deposit amount: Rs. ";

while (!(cin >> deposit) || deposit <= 0) {
    cout << "Invalid amount! Enter again: ";
    cin.clear();
    cin.ignore(numeric_limits<streamsize>::max(), '\n');
}

balance += deposit;

cout << "Rs. " << deposit << " deposited successfully.\n";
cout << "New Balance = Rs. " << balance << endl;

break;
}

case 3: {
    double withdraw;

    cout << "Enter withdrawal amount: Rs. ";

    while (!(cin >> withdraw) || withdraw <= 0) {
        cout << "Invalid amount! Enter again: ";
        cin.clear();
        cin.ignore(numeric_limits<streamsize>::max(), '\n');
    }

    if (withdraw > balance) {
        cout << "Insufficient Balance!\n";
    }
    else {
        balance -= withdraw;
        cout << "Rs. " << withdraw << " withdrawn successfully.\n";
        cout << "Remaining Balance = Rs. " << balance << endl;
    }

    break;
}

case 4:
    cout << "\nThank you for using our ATM.\n";
    break;

default:
    cout << "\nInvalid choice! Please select 1-4.\n";
}

} while (choice != 4);

return 0;
}
```

5. Guess the secret number

```
#include <iostream>
using namespace std;

int main(){
    int guess = 9;
    int num;

    do{
        cout <<"Enter a Secret Number: ";
        cin >> num;

        if(guess != num){
            cout <<"Wrong Answer! Try again: "<<endl;
        }
    }while(guess != num);

    cout <<"Congratulation..." <<endl;

    return 0;
}
```

6. Keep asking for password until correct

```
#include <iostream>
#include <string>
using namespace std;

int main(){
    string pass_code = "Sourabha@1973";
    string pass_word;

    do{
        cout <<"Enter your Password: ";
        cin >>pass_word;

        if (pass_word != pass_code){
            cout <<"Wrong Password! try again: ";
        }

    }while(pass_word != pass_code);

    cout<<"Access Granted!"<<endl;
```

```
    return 0;
}
```

7. Print multiplication table

```
#include <iostream>
using namespace std;

//Print multiplication table
int main(){
    int num;
    cout <<"Enter a Number for Multiplication Table: ";
    cin >>num;

    int i  = 1;

    do{
        cout << num <<" * " << i << " = " <<num * i <<endl;
        i++;
    }while(i <= 10);

    return 0;
}
```

8. Sum numbers until user enters 0

```
#include <iostream>
using namespace std;

int main (){
    double num1,sum = 0 ;

    do{
        cout <<"Enter number: ";
        cin >>num1 ;

        sum = sum + num1;

    }
    while(num1 != 0);

    cout <<"Sum is : " <<sum <<endl;
    cout <<"Thank You! " <<endl;

    return 0;
}
```

9.Find factorial

```
#include <iostream>
using namespace std;

//Find factorial
int main(){
    int num;
    cout <<"Enter a Number: ";
    cin >>num ;

    int fact = 1;

    do{
        fact = fact * num;
        num--;
    }while(num >= 1);

    cout <<"Factorial is " <<fact <<endl;

    return 0;
}
```

10.Repeat program until user chooses Exit

```
#include <iostream>
using namespace std;

int main() {

    int choice;

    do {

        cout << "\n===== MENU =====\n";
        cout << "1. Say Hello\n";
        cout << "2. Show Number\n";
        cout << "3. Exit\n";

        cout << "Enter Choice: ";
        cin >> choice;

        switch (choice) {
```

```
        case 1:
            cout << "Hello!";
            break;

        case 2:
            cout << "Number = 100";
            break;

        case 3:
            cout << "Goodbye!";
            break;

        default:
            cout << "Invalid Choice";
    }

} while (choice != 3);

return 0;
}
```